

VLSI · IECON 2026 CANDIDATE PAPER

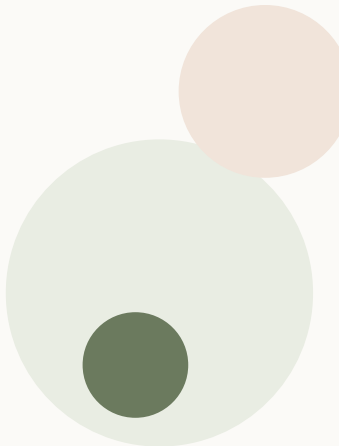
An Activity-Rate-Aware Hysteresis Controller for Adaptive Power Gating

Eliminating sleep-mode thrashing in SoC blocks — without machine learning

Joud Thaher · Layan Salem

Supervised by Dr. Khader Mohammed

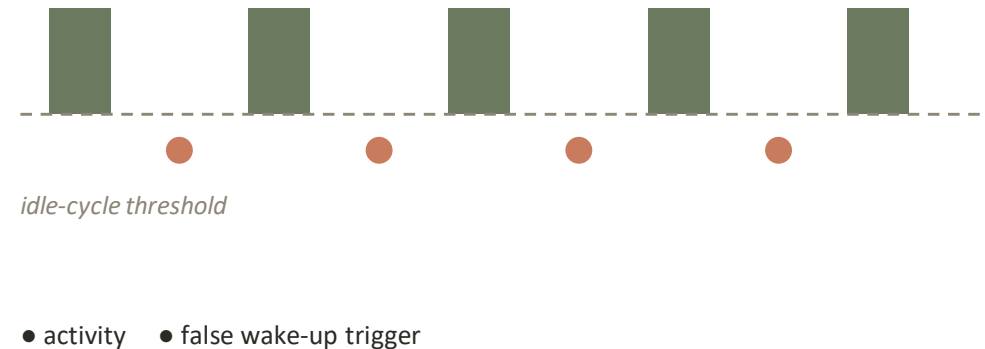
Department of Electrical and Computer Engineering — Birzeit University



Static thresholds can't tell idle apart from busy-with-gaps

- Leakage power now dominates total power as nodes shrink.
- Power gating helps — but most controllers sleep after a fixed idle-cycle count, regardless of why the gap occurred.
- Bursty traffic with short gaps above that threshold causes repeated sleep/wake cycles — thrashing.

A bursty workload, seen by a fixed threshold



Each wake-up costs latency and energy — savings from sleeping briefly can cost more than they save.

Gate sleep entry on idle count AND activity rate — not idle count alone



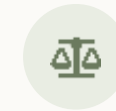
When recent activity is high, both thresholds widen — bursty traffic no longer reads as idle.



Comparator-only logic — no multipliers, no learned models



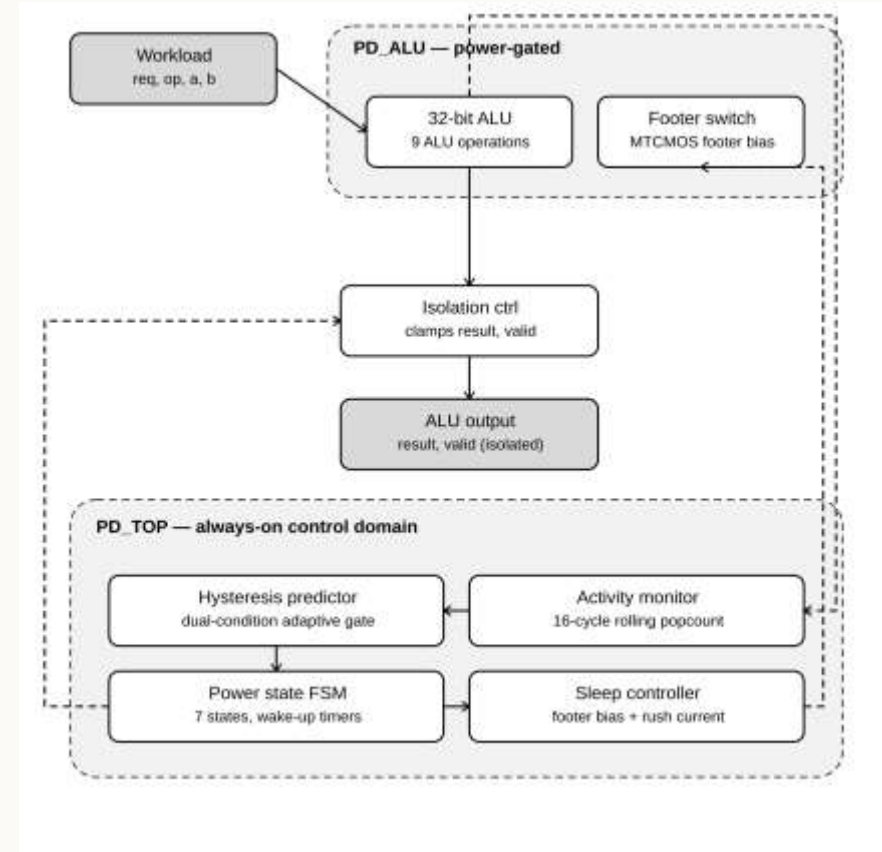
Adaptive threshold widening suppresses sleep during bursts



Fair, single-module-substitution comparison isolates the real variable

Two power domains, six modules, one substituted block

- PD_TOP — always-on control domain (monitor, predictor, FSM, sleep controller)
- PD_ALU — power-gated domain (32-bit ALU + MTCMOS footer switch)
- The hysteresis predictor is the controller's only novel block



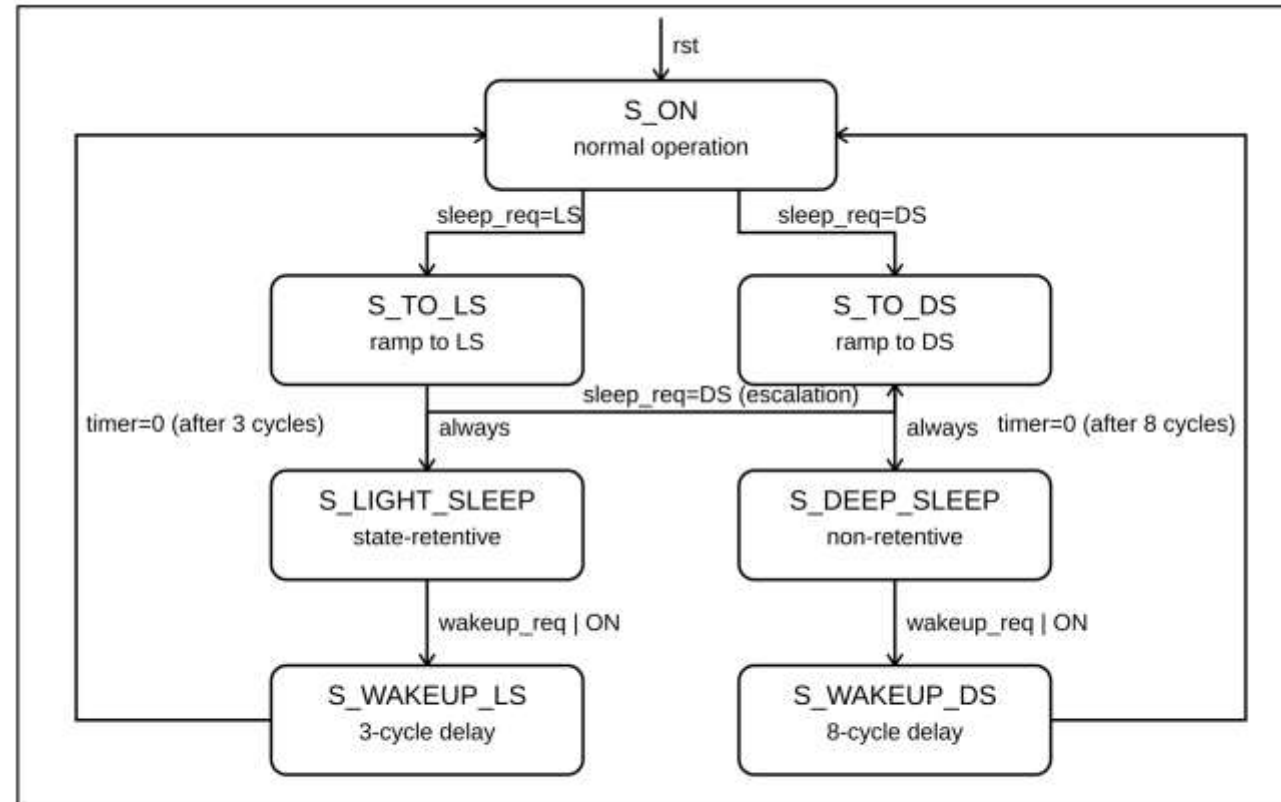
Seven states govern every transition

3

cycles · LS wake-up

8

cycles · DS wake-up



Self-transitions (remaining in the same state) omitted for clarity

One module swapped. Everything else identical.

Baseline

- 32-bit ALU
- Activity monitor
- Power-state FSM
- Sleep controller
- Isolation control
- UPF power intent

Fixed-threshold predictor

Proposed

- 32-bit ALU
- Activity monitor
- Power-state FSM
- Sleep controller
- Isolation control
- UPF power intent

Hysteresis predictor (novel)

only variable that differs

Three workloads, identical stimulus, both designs



BURST

Short bursts, gaps above the baseline's fixed threshold



SPARSE

Long, unambiguous idle periods

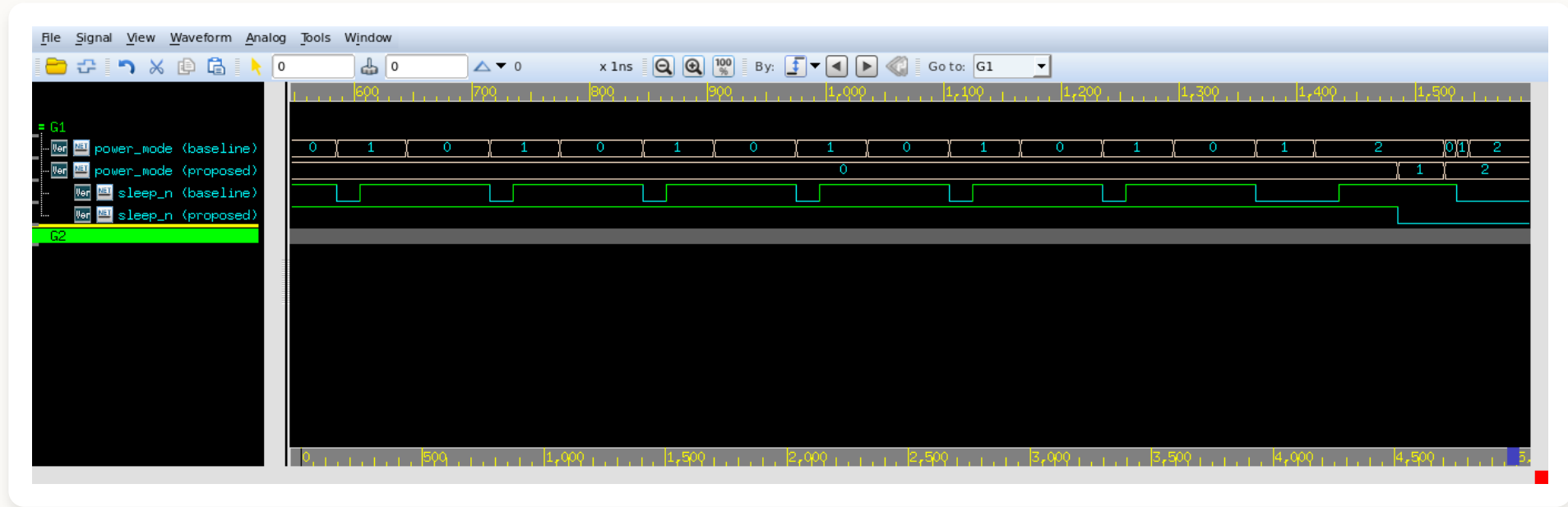


MIXED

Alternating dense and sparse phases

Synopsys VCS · Design Compiler · IC Compiler II · SAED14LVT 14 nm

Zero transitions vs. nineteen



BURST workload — power_mode and sleep_n, baseline vs. proposed

19 → 0

mode transitions

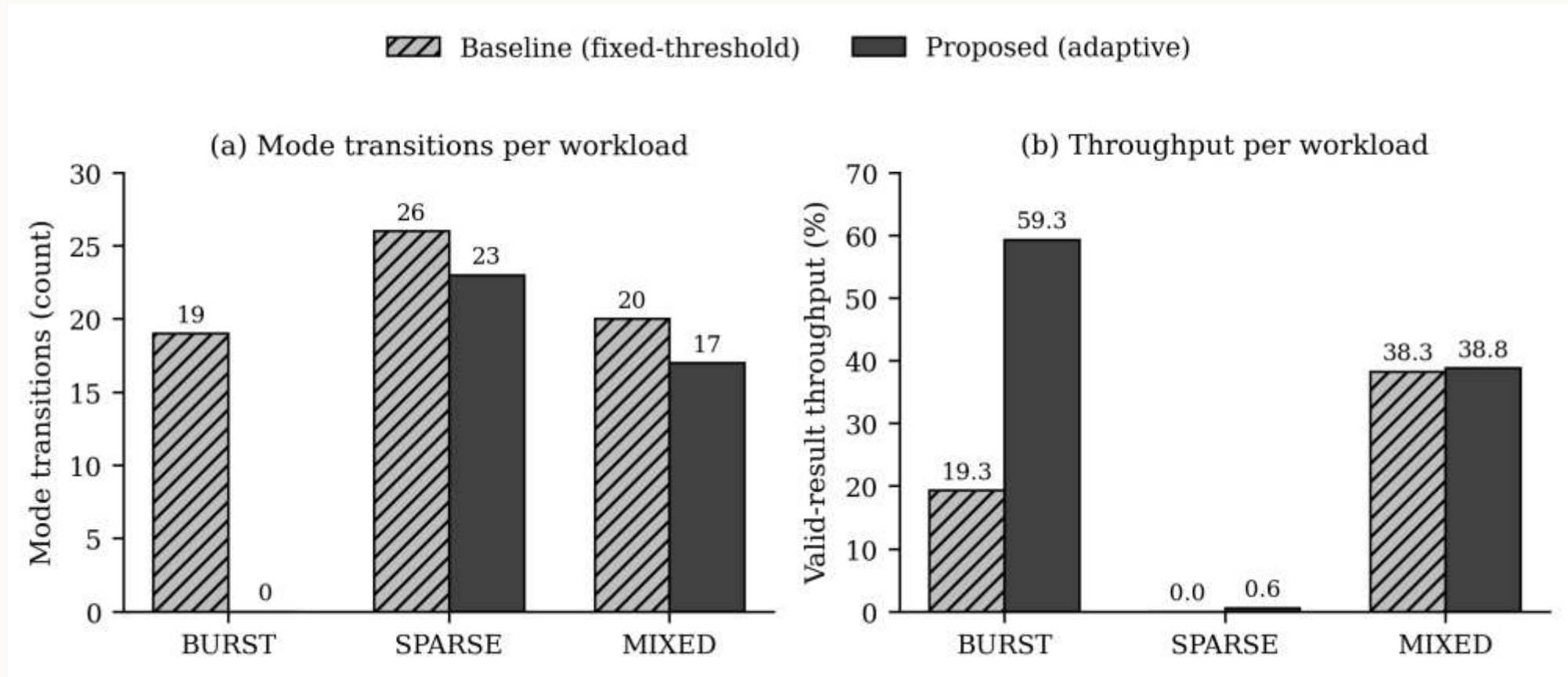
9 → 0

wake-up events

19.3% → 59.3%

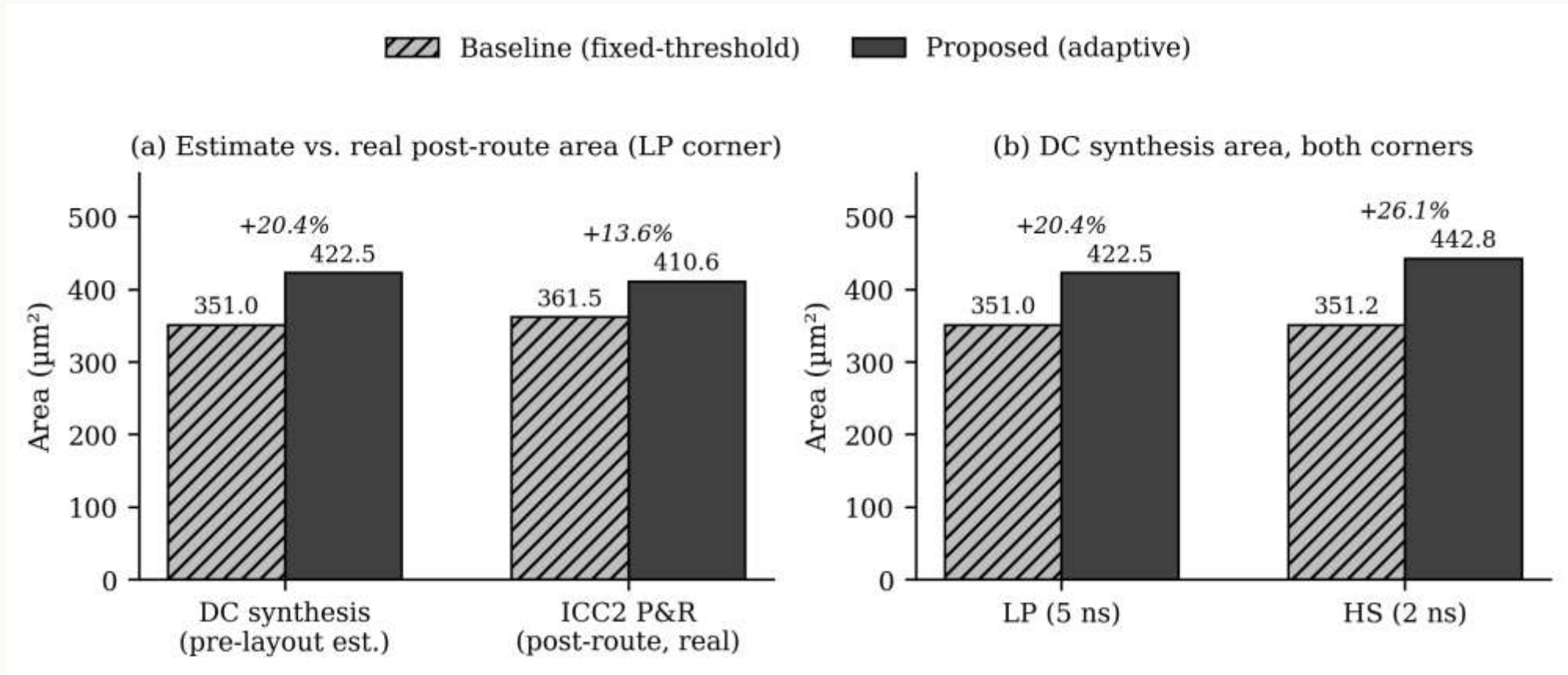
throughput

Stable under bursts. Equally efficient when truly idle.



Sparse workload: 33.4% vs. 31.6% leakage saving — within measurement noise, not a real penalty.

A bounded cost for a real behavioral win



+13.6% real post-route area · WNS = TNS = 0.0000 ns on both designs

Where this sits against prior work

Approach	Decision input	ML hardware?	Outcome
Agarwal et al. — multi-level gating	Idle-count only	No	Thrashes under bursts
Karthikeyan et al. — multi-core ML	Learned classifier	Yes (ML)	Sized for multi-core, not one block
This work	Idle-count + activity rate	No (comparators)	Bounded +13.6% area, zero thrashing

What this doesn't claim — yet



Energy comparisons use a derived metric (transitions × wake-up latency), not a first-principles joule measurement



Post-route results cover a single corner; multi-corner PVT analysis was outside this project's scope



Workloads are synthetic, designed to probe threshold-crossing behavior — not drawn from a real application trace

Future work: extend the activity-rate gate to other block types, multi-corner verification, and real application traces.



CONCLUSION

Comparator logic. Real silicon. One clear trade-off.

3.1×

throughput under burst traffic

13.6%

real post-route area overhead

0

timing slack — both designs

Results confirm the proposed design eliminates sleep-state thrashing while converging to baseline-level savings under genuinely idle conditions — at a real, bounded silicon cost.



Thank you.

Questions?

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